

Smart House Monitoring Platform

IoT Environmental Monitoring · Python / Streamlit · SQLite · Plotly Visualization

An interactive web platform that centralizes real-time household environmental monitoring — temperature, humidity, gas and light — into a single operator-oriented dashboard with historical data, customizable thresholds and multilingual support.

Python

Streamlit

Plotly

SQLite

IoT HMI

FR / EN / AR

3

Monitored zones

4

Sensor types

3

UI languages

Real-time

Data refresh

From scattered sensors to a unified monitoring platform

THE PROBLEM

Fragmented environmental data

Household environmental information — temperature, humidity, gas and light — is typically captured by separate devices and never centralized. There is no unified view, no history, no alert logic.

Consequences

- No global situation awareness
- No historical trend analysis
- Thresholds checked manually
- No operator-oriented visualization

THE OBJECTIVE

A single interactive platform

Build a Python-based web application that aggregates all sensor streams into a single dashboard, stores history in SQLite, and exposes configurable thresholds and alerts to the operator.

Integrated capabilities

- Real-time KPI cards
- Historical time-series
- Configurable thresholds
- Multilingual UI (FR / EN / AR)

Engineering approach

The platform is designed as a five-layer software system. Each layer has a single responsibility: acquiring sensor data, processing it in Python, persisting it to SQLite, exposing it through a Streamlit interface, and rendering it as interactive Plotly charts.

This separation keeps the application modular and testable.

Key features

- Real-time sensor KPI cards per room
- Interactive time-series (Plotly)
- Historical data persisted in SQLite
- User-configurable alert thresholds
- Color-coded room status (normal / alert)
- Multilingual interface: FR / EN / AR

Technology stack

Language	Python 3
Interface	Streamlit
Charts	Plotly
Storage	SQLite
i18n	FR / EN / AR

A five-layer monitoring pipeline

Each stage of the pipeline has a clearly separated role. Data flows unidirectionally from the physical sensors toward the interactive visualization layer, with persistence decoupled from display.



LAYER 01 Data Physical measurements from temperature, humidity, gas and light sensors distributed across rooms.	LAYER 02 Processing Python modules normalize, validate and compute derived values before storage or display.	LAYER 03 Storage SQLite provides lightweight, file-based persistence with structured tables for measurements and thresholds.	LAYER 04 Interface Streamlit renders the interactive HMI: KPI cards, room layout, detail panel, threshold sliders.	LAYER 05 Visualization Plotly renders interactive time-series with hover, zoom and dynamic refresh.
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DESIGN PRINCIPLE

Strict separation between data, logic, storage and presentation.

Each layer communicates with the next through a stable interface — the application can be evolved, tested or replaced layer by layer.

The operator interface and the data layer

Dashboard organization

The HMI is organized into four functional areas so that the operator can move from global overview to room-level detail in one click.

LAYOUT

01 Overview strip

Three time-series charts (temp / humidity / gas).

02 Room plan

Color-coded floor layout with per-zone status.

03 Detail panel

Focused room view with threshold slider and alert state.

04 KPI cards

Per-room summary with current values and alarm indicators.

SQLite data model

Two tables are sufficient to support both historical plotting and threshold logic. The schema is intentionally small and readable.

TABLE MEASUREMENTS

<code>id</code>	INTEGER PRIMARY KEY
<code>timestamp</code>	DATETIME
<code>room</code>	TEXT — Salon / Chambre / SdB
<code>sensor</code>	TEXT — temp / hum / gas / light
<code>value</code>	REAL

User interaction

- Dropdown selector to focus on a specific room and sensor
- Threshold sliders with immediate state re-evaluation
- Hover tooltips on Plotly charts for exact values
- Auto-refresh on a configurable cadence
- Language switch between FR / EN / AR

TABLE THRESHOLDS

<code>room</code>	TEXT PRIMARY KEY
<code>sensor</code>	TEXT
<code>min_value</code>	REAL
<code>max_value</code>	REAL

ALERT LOGIC

State derived from thresholds

If `value > max_value` $\Rightarrow$ room marked ALERT. State re-evaluated on every data refresh and threshold change.

Multilingual interface

FR

Français

Température · Humidité · Gaz ·
Seuil · État

EN

English

Temperature · Humidity · Gas ·
Threshold · Status

AR

All labels translated, RTL layout supported in the interface.

The dashboard in action

The full interface, rendered by Streamlit and Plotly, combines the three chart panels, the color-coded room plan, the detail panel with threshold slider, and the per-room KPI cards.



FIG. 01 Complete Streamlit dashboard — overview strip, room plan, detail panel and KPI cards

KPI CARDS

Per-room real-time values (temp / hum / lx) with color-coded status derived from thresholds.

INTERACTIVE CHARTS

Plotly time-series with hover tooltips, zoom and pan — rendered natively inside Streamlit.

TIME-SERIES

Continuous acquisition persisted in SQLite; historical window configurable by the operator.

DATA FILTERING

Room selector and threshold slider drive live re-evaluation and refresh of every downstream component.

STREAMLIT COMPONENTS

`st.columns` for layout · `st.selectbox` for room selection · `st.slider` for thresholds · `st.plotly_chart` for interactive rendering · `st.cache_data` for query optimization.

What this project proves

This project demonstrates the ability to design and ship a complete software system: from architectural decomposition to a working operator-oriented interface, with persistence, internationalization and interactive visualization.

01

Software Engineering

Python application development, modular architecture, clean separation between data, logic and presentation.

02

Industrial / IoT HMI

Operator-oriented interface design, real-time KPI visualization, interactive controls and alert logic.

03

Data Handling

SQLite schema design, historical time-series storage, Plotly-based interactive visualization.

04

Engineering Mindset

System architecture, human-machine interaction, multilingual delivery and separation of concerns.

Python · Streamlit · Plotly · SQLite · Multilingual HMI ;